The Impact of Central Bank Stock Purchases:
Evidence from Discontinuities in Policy Rules *

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Abstract

We trace the impact of central bank stock purchases by exploiting discontinuities in the Bank of Japan's policy rule, which triggers purchases when stock market index movements fall below a threshold. In normal times, the purchases raise the long-term interest rate while leaving no detectable impact on stock prices. After the introduction of yield curve control, which pegs the long-term yield at 0%, interest rates stop responding and stock prices rise robustly following the purchases. Although these results are inconsistent with the inelastic market hypothesis in its original form, general equilibrium extensions with endogenous interest rates explain our findings.

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1 Introduction

How does a financial flow into the stock market impact asset prices? While textbook models predict almost no effect, the inelastic stock market hypothesis recently proposed by Gabaix and Koijen (2021) argues for a large effect on stock prices. Answering this question goes a long way toward understanding the source of asset price volatility and the effectiveness of central bank asset purchase programs. In fact, as a new form of "quantitative easing," the Bank of Japan (henceforth BoJ) started to purchase stocks in 2010. At the end of 2020, the BoJ owned more than 6% of the stock market capitalization, becoming the largest owner of Japanese stocks worldwide. The BoJ explains that the primary goal of this extreme form of quantitative easing is to "reduce the risk premium," but its effectiveness is often subject to policy debates.

The reason why this question is difficult to answer is the endogeneity of financial flows. No hedge fund randomizes when trading financial assets, and the central bank intervenes in the financial market for a reason. For example, a hedge fund might put money into the stock market when it believes the stock market will perform well, or the BoJ might purchase stocks when the stock market performs poorly. Using these kinds of variations in financial flows will give misleading answers to the question because of reverse causality.

In this paper, we address this challenge by exploiting discontinuities in the BoJ's policy rules. Although the BoJ has never made its policy rule public, it is widely known that the BoJ tends to purchase stocks precisely on the day when the changes in the stock market index in the morning session fall below a certain threshold. By comparing days when the movements in a stock market index fall slightly below the threshold and slightly above, the discontinuous increase in inflow into the stock market can be viewed as orthogonal to the underlying economic fundamentals. A difficulty in implementing the standard regression discontinuity design in our setup is that the threshold is not necessarily known, as the BoJ has never made the policy rule public. We overcome this problem by estimating the threshold relying on the econometric literature on regression discontinuity design with unknown cutoffs (Porter and Yu, 2015). Indeed, we find a striking discontinuity in the likelihood of the intervention around the estimated cutoffs.

Exploiting the discontinuity in the policy rules, we first show that flows into stock markets have a large causal impact on *both* the stock prices and the long-term government bond interest rates. In response to an average-sized intervention, which amounts to 0.01% of stock market capitalization, our estimates indicate that the stock prices rise by around 0.1-0.4% following the intervention. Perhaps more surprisingly, the 10-year Japanese

government bond (JGB) yield also rises by around 0.5 basis points (b.p.). Although the standard errors increase, these results appear to be persistent, lasting at least several days.

We then argue that the above results mask a stark underlying heterogeneity that depends on the presence of another unconventional monetary policy, yield curve control (YCC). In the middle of 2016, the BoJ pegged the 10-year JGB yield at 0%. Since then, long-term interest rates have stabilized at around 0%. Given this, it is natural to expect the response of long-term interest rates to the financial flows to be different before and after the introduction of the YCC, and we find that this is indeed the case.

Specifically, before the introduction of the YCC, in response to the BoJ's stock market purchases, the long-term interest rates rise sharply and persistently following the intervention, while leaving virtually no detectable impact on the stock price. Quantitatively, we find that long-term interest rates rise by 1.0-1.5 b.p. in response to an average-sized intervention. In contrast, we cannot reject the null that the BoJ's stock purchases had no impact on stock prices in the following days, although the standard errors are large.

After the introduction of the YCC, the long-term interest rate stops responding entirely, and instead, stock prices rise sharply and persistently in response to the BoJ's stock purchases. The effect on the long-term interest rate is precisely estimated at zero, suggesting that yield curve control successfully stabilizes the long-term interest rate. A typical intervention raises stock prices by around 0.2-0.4%, and this increase persists for several days after the intervention.

In the final part of the paper, we argue that our results are consistent with models of inelastic asset demand in which the scope for interest-rate adjustment determines whether stock purchases raise stock prices or interest rates (Duarte et al., 2025; Caballero et al., 2024). These models extend the inelastic market hypothesis of Gabaix and Koijen (2021) by endogenizing the interest rate. In its original form, the model of Gabaix and Koijen (2021) features an exogenous interest rate and therefore cannot explain the strong interest rate response that we see before YCC.

Once the interest rate is endogenized through goods market clearing, as in Duarte et al. (2025), the inflow into the stock market raises the interest rate without changing the stock price. In the model, an inflow into the stock market puts upward pressure on stock prices. If stock prices were to rise, this would induce a stock market wealth effect on consumption. However, the rise in consumption is incompatible with goods market clearing under exogenous output. In fact, any movement in the stock price would violate goods market clearing, so the stock price must remain unchanged. What equilibrates the excess demand for stocks induced by the central bank's stock purchases is the interest rate. The interest rate rises, the demand for stocks falls, and the stock market clears,

consistent with the pattern we find before YCC.

When the central bank fixes the interest rate, as in Caballero et al. (2024), the inflow into the stock market is now entirely absorbed by the stock price, consistent with the pattern we find after YCC. The higher stock price raises consumption, and the goods market clears through an increase in output, as output is demand-determined. The stock market clears through the higher stock price, instead of the interest rate. We formalize these mechanisms using a simple theoretical framework in Appendix G.

Quantitatively, the magnitudes of price responses that we document suggest that asset demand is substantially inelastic. After YCC, an inflow of $1 into the stock market raises aggregate stock market value by $20–$30, compared to the $5 multiplier in Gabaix and Koijen (2021). Before YCC, an inflow of 1% of stock market capitalization raises the interest rate by 1.0–1.5 percentage points, which is a sizable effect. Taken together, our findings suggest that a small stock-price response need not imply elastic asset demand, because inelasticity can instead manifest itself in a large interest-rate response when interest rates can adjust. If the interest rate does not adjust, the stock price has to absorb the inflow into the stock market.

Related Literature

We build most directly on the pioneering work by Gabaix and Koijen (2021) in assessing how a flow from bonds to stocks impacts the financial market. Gabaix and Koijen (2021) and a growing number of studies (e.g., Da et al., 2018; Hartzmark and Solomon, 2023; Li et al., 2021; Parker et al., 2023) estimate how flows from bonds to stocks affect stock prices using various identification strategies. Our contributions to this literature are twofold. First, our empirical results reject the inelastic market hypothesis in its narrow form, but give strong support to it when augmented with the general equilibrium interest rate adjustment (Duarte et al., 2025; Caballero et al., 2024). Second, we propose a novel identification strategy relying on discontinuities in the Bank of Japan's policy rules. An advantage of our approach is that it transparently points toward the source of identification and the underlying assumptions.

To the best of our knowledge, our paper is the first to identify the aggregate effect of the Bank of Japan's stock purchases. Many studies exploit the difference in weights in the BoJ's purchase basket to identify relative price impacts on stocks with larger versus smaller weights (Barbon and Gianinazzi, 2019; Charoenwong et al., 2021; Harada and Okimoto, 2021; Adachi et al., 2021; Katagiri et al., 2022). In contrast, our empirical strategy allows us to focus on the aggregate effect. In this regard, various studies (Shirota, 2018;

Fukuda and Tanaka, 2022; Chung, 2020; Hattori and Yoshida, 2023; Ichiue, 2024) assume selection on observables and use the unexplained policy variation that remains after conditioning on observables to identify the aggregate effect. However, it is difficult to rule out the presence of unobservables that simultaneously affect the financial market performance and the likelihood of BoJ intervention. Our identification assumptions, which only require continuity of economic fundamentals with respect to stock price changes in the morning session, are substantially weaker than those of these studies.

More broadly, we contribute to the large literature studying the effect of central bank asset purchases, so-called "quantitative easing" (e.g., Krishnamurthy and Vissing-Jorgensen, 2011; Chodorow-Reich, 2014; Droste et al., 2021; Selgrad, 2023). These studies have focused on central bank purchases of long-term government bonds or mortgage-backed securities, which are swaps between one type of bond (e.g., long-term bonds) and another (e.g., reserves). Our focus is conceptually different from these because central bank stock purchases are swaps between stocks and bonds.

While there are many studies that isolate quasi-experimental variation in monetary policy, our approach is unique in exploiting the discontinuity in the policy rule. The closest to our approach is the one in Kuersteiner, Phillips, and Villamizar-Villegas (2018), who also use a discontinuous policy rule to investigate the effectiveness of sterilized foreign exchange interventions in Colombia. Our approach differs not only in terms of the empirical context but also in methodology, as we use the technique of regression discontinuity with unknown discontinuity points (Porter and Yu, 2015).

2 Data

Our primary goal is to measure the impact of an inflow into the stock market induced by central bank stock purchases. Our estimation sample runs from December 15, 2010, the date of the BoJ's first ETF purchase, through the end of 2020. The program was announced in October 2010. The BoJ purchases exchange-traded funds (ETFs) indexed to stock market indexes in Japan. We do not distinguish between ETFs and stocks and use the terms interchangeably. We obtain the dates and amounts of stock purchases for each of the BoJ's interventions from the BoJ website. Figure 1A shows the cumulative amount of ETF purchases over time. The BoJ started the stock market purchases in December 2010 as a new form of quantitative easing. By the end of 2020, the BoJ held over 6% of the stock market capitalization in Japan. The average size of each intervention is roughly 500 million USD, corresponding to 0.01% of the stock market capitalization. The total number of interventions during our sample period is 668. The amount of ETF

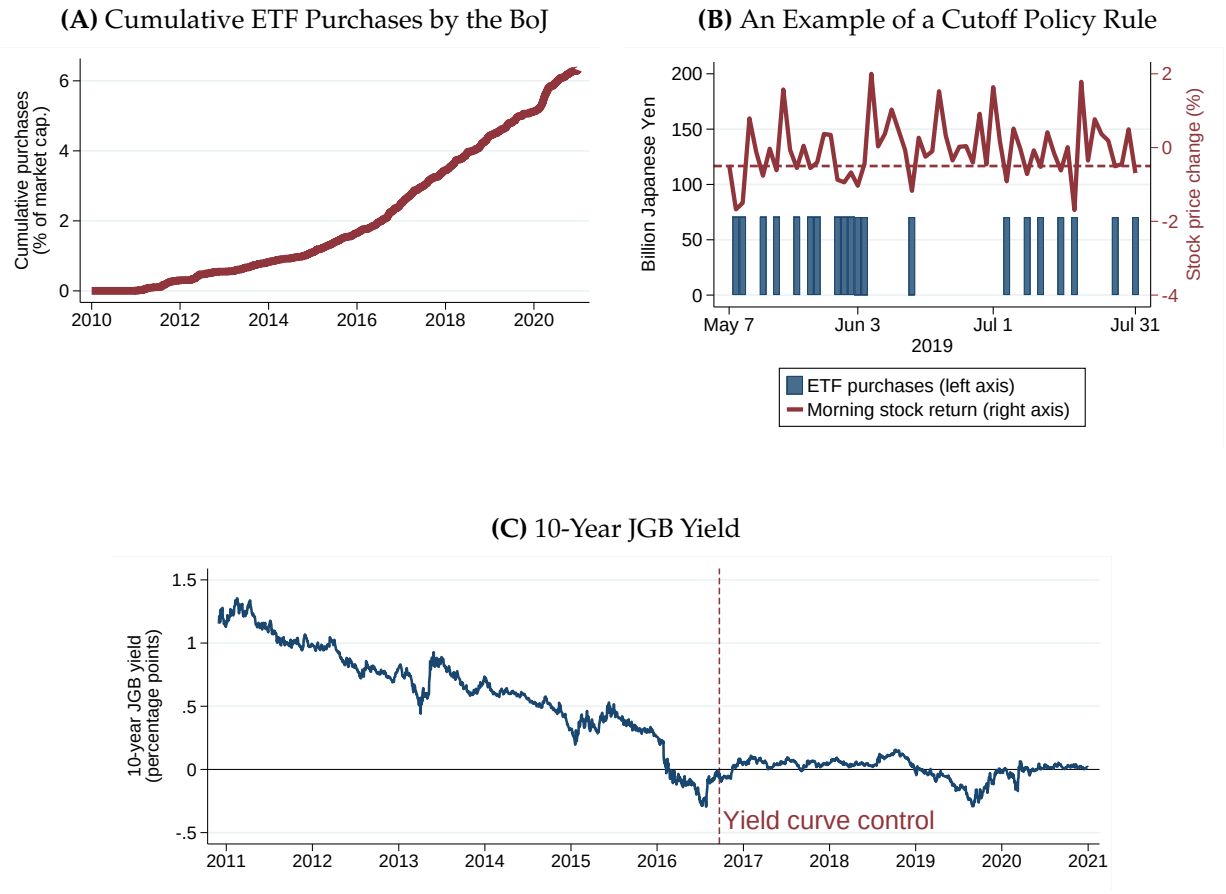


Figure 1: Unconventional Monetary Policies by the Bank of Japan

Notes: Panel A plots the cumulative amount of stock purchases by the BoJ from 2010 to 2020 as a fraction of stock market capitalization. Panel B illustrates the cut-off policy rule by showing the percentage changes in TOPIX and the BoJ (Bank of Japan) purchase amount for each day from May 2019 to July 2019. The solid red line shows the TOPIX changes in the morning session, and the dashed red line is the estimated cutoff of 0.5%. The bar shows the amount of purchases for each intervention in billions of Japanese yen (approximately 10 million US dollars). Panel C shows the path of the 10-year JGB (Japanese government bond) yield over time, where the red vertical dashed line (September 21, 2016) denotes the start of the yield curve control.

purchases is normalized by the stock market capitalization, which we obtained from the Japan Exchange Group Data Cloud.

The BoJ publishes the amount of ETF purchases on the morning of the day following the intervention. Based on the trading volume, it is widely believed that the BoJ submits an order at lunchtime, although the BoJ has never made this practice public (Harada and Okimoto, 2021). Therefore, investors potentially face uncertainty about whether the large inflow into the stock market reflects the BoJ's intervention or other factors on the day of the intervention. For this reason, we prefer to use our empirical estimates on the next day as our benchmark estimates.

To measure the response of the stock market, we use tick-by-tick data on the Tokyo Stock Price Index (henceforth, TOPIX), which is the index of the Tokyo Stock Exchange in Japan, tracking all domestic companies in the exchange's first section. We obtain these data from the Japan Exchange Group Data Cloud. To measure the response of the long-term interest rate, we use tick-by-tick Japanese government bond yield data, which we obtained from Refinitiv Japan. Since some observations are missing in the Refinitiv data, we supplement those missing observations with the hourly data from Bloomberg. We aggregate the tick-by-tick data to an hourly frequency by averaging observations within one-hour intervals.[1]

3 Empirical Strategy

We consider the following econometric model:

$$\Delta y_{t+l,h} = \beta_{l,h} \times ETF_t + \Gamma'_{l,h}\mathbf{X}_t + \epsilon_{t+l,h}, \tag{1}$$

where $\Delta y_{t+l,h} \equiv y_{t+l,h} - y_{t,0}$ is the change in the variable y (e.g., the log of stock prices) from the end of the morning session ($h = 0$) on day t to time h on day $t + l$, ETF_t is the amount of stock market purchases by the BoJ relative to the stock market capitalization of Japan, $\mathbf{X}_t$ is the vector of controls, and $\epsilon_{t+l,h}$ contains the unmodeled determinants of the outcome variable. We are interested in estimating $\beta_{l,h}$, which measures the effect of the central bank's stock purchases on the outcome observed at time h, l days after the purchase. We choose this simple linear model for expositional purposes. In Appendix A.1, we consider a nonlinear version of (1) and present a more technical interpretation of

[1]For example, "10 AM" observation averages the ticks recorded between 10 AM and 11 AM. Before November 21, 2011, the morning session closes at 11 AM, so the "11AM" observation averages the ticks recorded at the morning close. From that date onward, the morning session closes at 11:30 AM, so "11AM" averages observations from 11 AM through the morning close.

the estimated parameter as in Angrist and Imbens (1995). In the baseline specification, we do not include any controls $\mathbf{X}_t$, but we show robustness to adding various controls in Section 4.3.

An obvious concern with estimating equation (1) by OLS is reverse causality. For example, one might expect that the BoJ is more likely to intervene when the stock market is expected to perform poorly, which leads to a downward bias in the OLS estimates of $\beta_{l,h}$. Alternatively, if the BoJ intervenes during periods of high risk premia (so the subsequent return is high), the OLS estimates are biased upward.

To solve this endogeneity problem, we propose an identification strategy based on a regression discontinuity design, which builds on the observation that the BoJ's intervention appeared to follow a cut-off rule. It has been widely argued in the media that the BoJ seemed to intervene on days when movements in the value of TOPIX fell below a certain threshold in the morning session. For example, the Financial Times writes that "the central bank has tended to step in whenever the TOPIX index has lost more than 0.5 percent in the morning session."[2] In fact, Figure 1B shows that from May to July 2019, the BoJ followed a strict rule to intervene when the stock market index fell by more than 0.5% in the morning session. The BoJ intervenes when the index falls slightly below the 0.5% threshold, while it does not intervene when the index falls slightly above the threshold.

Suppose for the moment that such a cut-off is known. Then, we can apply a standard regression discontinuity design. Formally, we assume that the policy rule has the following form:

$$ETF_t = ETF_{-,t}(\Delta p_t)\mathbb{I}(\Delta p_t < c_t) + ETF_{+,t}(\Delta p_t)\mathbb{I}(\Delta p_t \geq c_t), \tag{2}$$

where Δp_t is the log-change in the TOPIX value in the morning, c_t is the cut-off, and $ETF_{-,t}$ and $ETF_{+,t}$ are some random functions of Δp_t that represent different policy rules depending on whether Δp_t is above or below the cutoff. We assume (i) $\mathbb{E}[\epsilon_{t+l,h}|\Delta p_t, \mathbf{X_t}]$ is continuous at $\Delta p_t = c_t$, (ii) $\lim_{\Delta p \uparrow c_t} \mathbb{E}[ETF_t|\Delta p_t = \Delta p, \mathbf{X_t}]$ and $\lim_{\Delta p \downarrow c_t} \mathbb{E}[ETF_t|\Delta p_t = \Delta p, \mathbf{X_t}]$ exist, and (iii) $\lim_{\Delta p \uparrow c_t} \mathbb{E}[ETF_t|\Delta p_t = \Delta p, \mathbf{X_t}] \neq \lim_{\Delta p \downarrow c_t} \mathbb{E}[ETF_t|\Delta p_t = \Delta p, \mathbf{X_t}]$. Under these assumptions, it follows that

$$\frac{\lim_{\Delta p \uparrow c_t} \mathbb{E}[\Delta y_{t+l,h}|\Delta p_t = \Delta p, \mathbf{X_t}] - \lim_{\Delta p \downarrow c_t} \mathbb{E}[\Delta y_{t+l,h}|\Delta p_t = \Delta p, \mathbf{X_t}]}{\lim_{\Delta p \uparrow c_t} \mathbb{E}[ETF_t|\Delta p_t = \Delta p, \mathbf{X_t}] - \lim_{\Delta p \downarrow c_t} \mathbb{E}[ETF_t|\Delta p_t = \Delta p, \mathbf{X_t}]} = \beta_{l,h}. \tag{3}$$

As recommended by Hahn, Todd, and Van der Klaauw (2001) and Porter (2003), we can devise local linear regression estimators for the left-hand side to obtain an estimate of $\beta_{l,h}$.

[2]Financial Times, "Bank of Japan backs away from ETF buying scheme," (March 23, 2021), https://www.ft.com/content/a654d1c9-7126-4587-8de6-ed15f567455f.

We use the optimal bandwidth proposed by Calonico, Cattaneo, and Titiunik (2014) and estimate $\beta_{l,h}$ using two-stage least squares and report Newey-West standard errors.

The difficulty in implementing the above approach, however, is that the cut-off is not necessarily known. While the cutoffs utilized by the BoJ were apparently known to the public in some periods, they were not known in other periods. In order to formally detect the cutoffs utilized by the BoJ, we estimate the cut-off with the presumption that the BoJ follows a cut-off rule, using the approach proposed by Porter and Yu (2015). They develop a method to estimate the discontinuity point and show that there is no loss of efficiency with the regression discontinuity estimator using the estimated cutoff. In implementing this approach, we proceed as follows. We first split the sample period to allow for time variation in the policy rule. We assume the cut-off is constant within the sample split. Then, in each sample split, we consider a set of possible cutoffs, $\mathbb{C} \equiv \{\bar{c}_1, \bar{c}_2, \ldots, \bar{c}_K\}$. For each $\bar{c} \in \mathbb{C}$, we estimate the jump in $\Pr_t(ETF_t > 0|\Delta p)$ around $\bar{c}$, which is

$$J_t(\bar{c}) \equiv \lim_{\Delta p \downarrow \bar{c}} \Pr_t(ETF_t > 0|\Delta p) - \lim_{\Delta p \uparrow \bar{c}} \Pr_t(ETF_t > 0|\Delta p).$$

We select $\bar{c}$ that maximizes the square of the jump, $J_t^2(\bar{c})$: $c_t^* \in \arg\max_{\bar{c} \in \mathbb{C}} J_t^2(\bar{c})$. We describe the details of the estimation procedure in Appendix A.2.

We implement the above approach with the following specifications. First, we split the sample period based on the BoJ's announcements regarding ETF purchases. The BoJ made six major announcements that changed the target amount of ETF purchases on April 4, 2013, October 31, 2014, December 18, 2015, July 29, 2016, July 31, 2018, and March 16, 2020. We further divide each period between two announcements based on whether the TOPIX close-to-close return is negative on each of the preceding two trading days. We make this choice based on widespread claims in the media,[3] and we have indeed found that it has strong explanatory power. Second, we consider the set of potential cutoffs ranging from -1% to 0% at 0.05% intervals. We estimate the jump in $\Pr_t(ETF_t > 0|\Delta p)$ around the potential cutoffs using local linear regressions with a fixed bandwidth of one percentage point on each side of the candidate cutoff.

[3]See, for example, the Bloomberg article "The BoJ's ETF Purchase Conditions Likely to Ease if Stocks Continue to Fall" (written in Japanese) (https://www.bloomberg.co.jp/news/articles/2020-07-22/-0-3-kcwteezj).

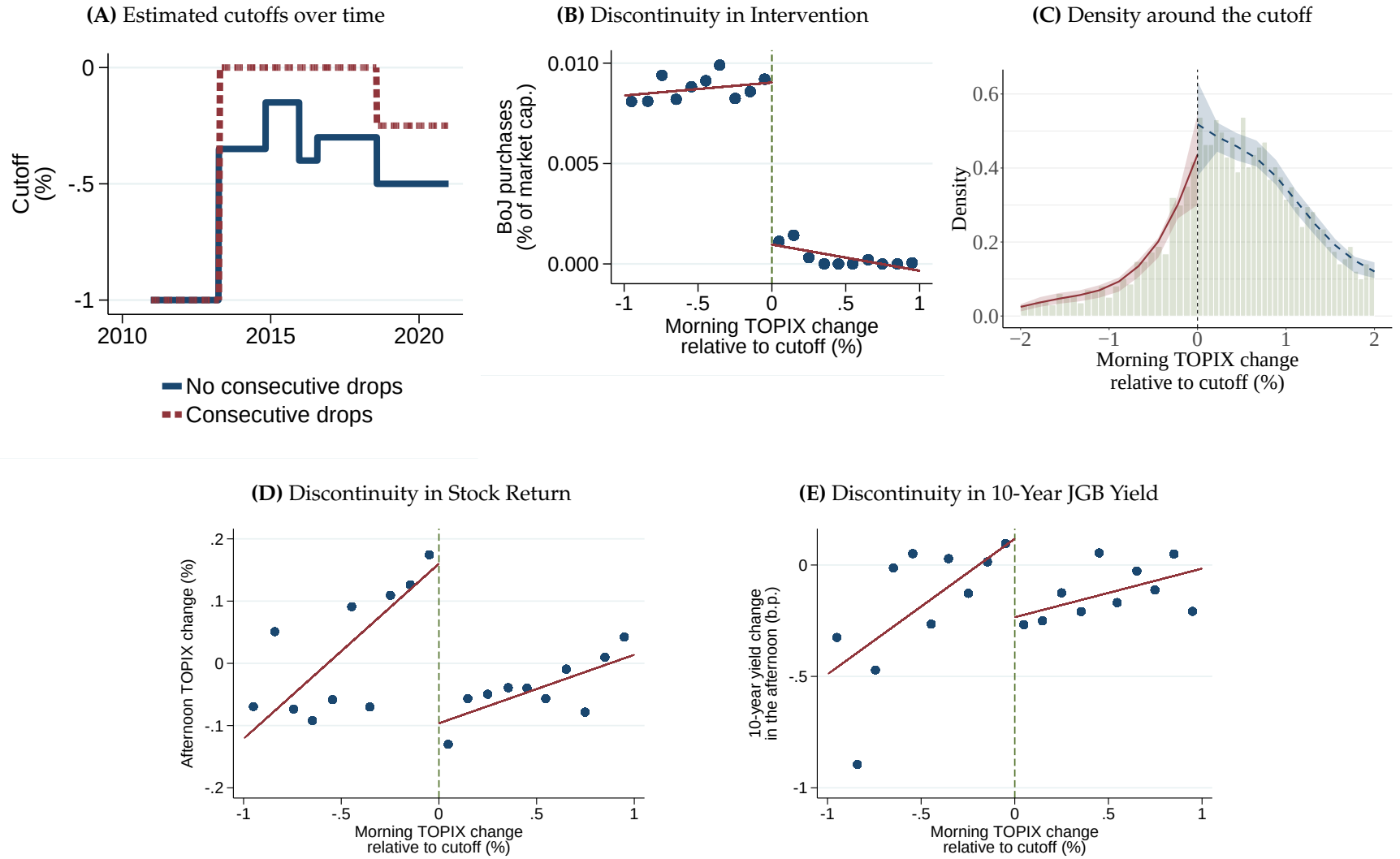


Figure 2: Discontinuities in the Policy Rule and Financial Market Outcomes

Notes: Panel A plots the path of estimated cutoffs over our sample period when there are no consecutive drops in the past two days (blue solid line) and when there are consecutive drops in the past two days (red dashed line). Panel B shows the discontinuity in BoJ stock purchases, measured as a percentage of market capitalization, within one percentage point of the estimated cutoff. Each dot plots the mean purchase amount against the mean morning TOPIX change relative to the cutoff within a bin, using ten equal-width bins on each side of the cutoff. The red lines show separate linear fits on each side. Panel C shows the histogram and the density of changes in TOPIX relative to the cutoff. The shaded area is the 95% confidence interval. We use the local polynomial density estimator of Cattaneo et al. (2020) with order 2. Panel D plots mean afternoon log changes in TOPIX (from 11 AM to 3 PM), expressed as percentages, against mean morning TOPIX changes relative to the cutoff, using ten equal-width bins on each side of the cutoff. The red lines show separate linear fits on each side. Panel E is constructed analogously, with the vertical axis showing the mean afternoon change in the 10-year JGB yield in basis points (b.p.).

Figure 2A shows the path of estimated cutoffs. The estimated cutoffs align well with the widely held consensus. During 2010-2013, it was widely believed that the BoJ followed a so-called "1% rule", in which the BoJ buys ETFs whenever the TOPIX falls by more than 1% in the morning session,[4] and our estimates confirm this view. Since April 2013, the BoJ appears to use different cutoffs depending on whether the daily change in the TOPIX has been negative for the past two consecutive days. Specifically, when the index has declined for two consecutive days, our estimates show a clear pattern in which the BoJ intervenes whenever the TOPIX falls below 0% in the morning session, confirming the claim in the media described earlier. Since July 31, 2018, the cutoffs appear to be 0.5% when there is no consecutive fall in the past two days, which is again consistent with the so-called "0.5% rule."

Figure 2B shows the binned scatter plot of the size of the BoJ intervention against changes in the TOPIX in the morning session of the same day relative to the cutoffs. We confirm that there is a discrete jump in the size of the BoJ interventions around zero. The implied jump in the full sample is 0.0083% of the market capitalization with a standard error of 0.0005%. The Cragg-Donald F-statistic is 1821, and the Kleibergen-Paap F-statistic is 261, eliminating weak identification concerns. This discontinuity comes from the discontinuity in the likelihood of intervention: the probability increases by 86.4 percentage points when moving from just above to just below the cutoff, with a standard error of 2.4 percentage points. Importantly, we find strong evidence of discontinuity in every sample split.[5]

A natural concern for a discontinuity-based research design is that the event of the index falling below the cutoff may not be random. For example, to the extent that the policy rule is widely known among investors, the optimal responses of investors may induce bunching of stock price changes in the morning session just below the cutoff. Under such manipulation of the stock price changes, the regression discontinuity estimator lacks causal interpretation, as explained in McCrary (2008).

Figure 2C plots the density of stock price changes in the morning session relative to the estimated cutoff. Under manipulation of stock price changes, we would expect bunching in the density on either side of the cutoff. However, we do not observe such bunching. Furthermore, Appendix A.3 formally tests for the presence of manipulation using the methodology proposed by Cattaneo et al. (2020). We do not find evidence of

[4]For example, Nikkei Asia writes "the BoJ was widely thought to be following an unwritten rule, dubbed the 1% rule: it would buy ETFs when the Topix index of all issues on the first section of the Tokyo Stock Exchange fell more than 1% in the morning session." (https://asia.nikkei.com/Business/Finance/BOJ-steps-up-REIT-buying-scales-back-ETF-purchases)

[5]We report the discontinuity for each sample split in Appendix A.2 and Figure B.1.

manipulation. This likely reflects the presence of uninformed ("noise") traders in the financial market, making it difficult for investors to manipulate stock prices.

4 Empirical Results

Armed with the cutoff estimates, we implement the regression discontinuity design to assess the impact of the BoJ's ETF purchases on the financial market.

4.1 Homogeneous Effect for the Entire Sample Period

Figure 2D shows the results for stock price changes. Specifically, it reports the binned scatter plot of changes in the TOPIX in the afternoon (from 11AM to 3PM) against changes in the TOPIX in the morning relative to the estimated cutoff. The figure shows that the stock returns were around 0.2 p.p. higher when the TOPIX fell slightly below the cutoff in the morning than when it fell slightly above the cutoff. Since the BoJ's intervention was likely to occur on days when the TOPIX fell below the threshold, and since the BoJ submitted the order to purchase ETFs during the lunch break, this suggests that the BoJ's intervention had a large impact on stock prices within the day. The magnitude is large considering that the BoJ purchased on average around 0.01% of market capitalization in each intervention.

Figure 2E focuses on the 10-year Japanese government bond yield as an outcome variable. Perhaps surprisingly, we also see a discontinuity in the long-term interest rate. The long-term interest rate is 0.4 basis points higher on the left side of the cutoff than on the right side. These within-day discontinuities motivate the dynamic specification in (1), which scales the price responses by intervention size and traces the dynamic responses.

Figures 3A and 3B plot the impulse response functions of stock prices and the 10-year JGB yield. Formally, we plot estimates of $\beta_{l,h}$ in equation (1) for each l and h, where l represents the number of days since the intervention and h represents the time in hours. In Figure 3A, we see an immediate and large stock price response in the afternoon of the intervention. It implies that a stock purchase of 0.01% of market capitalization, the typical size of an intervention, increases the stock price by 0.3-0.4%, which is statistically and economically significant. Over the subsequent three trading days, the point estimates are smaller at around 0.1-0.2%, but remain economically significant, although the confidence intervals widen. Reassuringly, we do not find any evidence of pre-trends, which is consistent with our identifying assumptions.[6]

[6]Appendix Figure B.6 looks at the pre-trends with various bandwidths further back and finds no

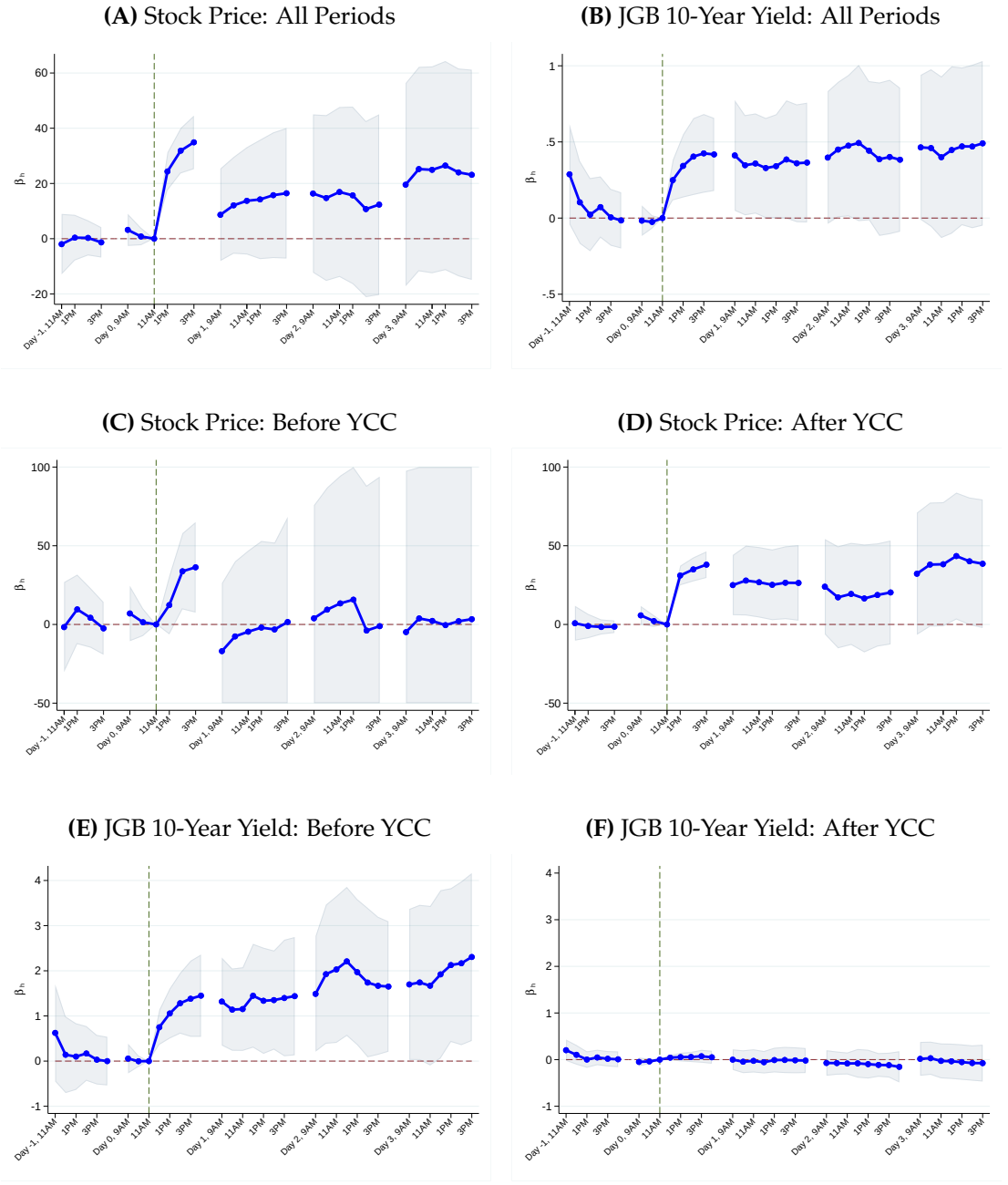


Figure 3: The Impact on Stock Prices and Long-Term Interest Rates

Notes: Panels A and B show the impulse response functions of stock prices and the 10-year JGB yield by plotting the coefficient $\beta_{l,h}$ in equation (1). The stock price coefficient measures the log changes in stock prices in response to stock purchases of 1% of market capitalization, and the yield coefficient measures the percentage point changes in the yield in response to stock purchases of 1% of market capitalization. Panels C and D show the impulse response of stock prices separately estimated before and after yield curve control, which is analogous to Panel A. Panels E and F show the impulse response of the 10-year JGB yield separately estimated before and after yield curve control, which is analogous to Panel B. In all figures, the shaded areas represent 90% confidence intervals, which account for heteroskedasticity and autocorrelation.

Figure 3B shows that the 10-year JGB yield also rises sharply after the intervention. Moreover, the effect appears to be quite persistent, and it remains statistically significant even several days after the intervention. The magnitude is again substantial. In response to a typical purchase size (0.01% of market capitalization), the 10-year JGB yield rises by around 0.4-0.5 basis points.

4.2 Heterogeneous Effects and Yield Curve Control

Figure 3B shows that the stock market purchases were accompanied by a rise in the long-term interest rate. In standard theoretical models, the rise in interest rates puts downward pressure on stock prices. For this reason, we expect that the ability of the interest rate to respond will be a critical determinant of the stock price responses to central bank stock purchases.

Another unconventional policy of the BoJ, the so-called "yield curve control," provides an ideal laboratory to explore this hypothesis. On September 21, 2016, the BoJ introduced an explicit target for the 10-year Japanese government bond yield at 0%. Figure 1C indeed shows that the long-term rate has stabilized at around 0% since the introduction of yield curve control. The daily standard deviation of the long-term rate is 0.37% before the introduction of yield curve control, but it falls to 0.08% after the introduction. If the BoJ does its best to stabilize the long-term interest rate at 0%, we would expect to see a much smaller response of the long-term rate in response to the stock purchases. To test this, we split our sample into periods before and after the introduction of yield curve control and rerun our analysis.

Figures 3C–3F show the main results of this paper. Figures 3C and 3D show the impulse response of stock prices before and after the introduction of yield curve control, respectively. Before the introduction of yield curve control, we find no evidence that the stock market responded positively beyond one day to the BoJ intervention, although the standard error is large. In stark contrast, stock prices rise persistently and in a statistically significant manner under yield curve control. Quantitatively, a purchase of 0.01% of stock market capitalization, the typical size of an intervention, causes around a 0.2-0.3% increase in stock prices several days after the intervention.

Figures 3E and 3F explain why. Figure 3E shows that the long-term interest rate responds positively before yield curve control. Quantitatively, a purchase of 0.01% of stock market capitalization by the BoJ causes around a one-basis-point increase in long-term rates, and the effect is statistically significant. However, under yield curve control,

evidence of pre-trends.

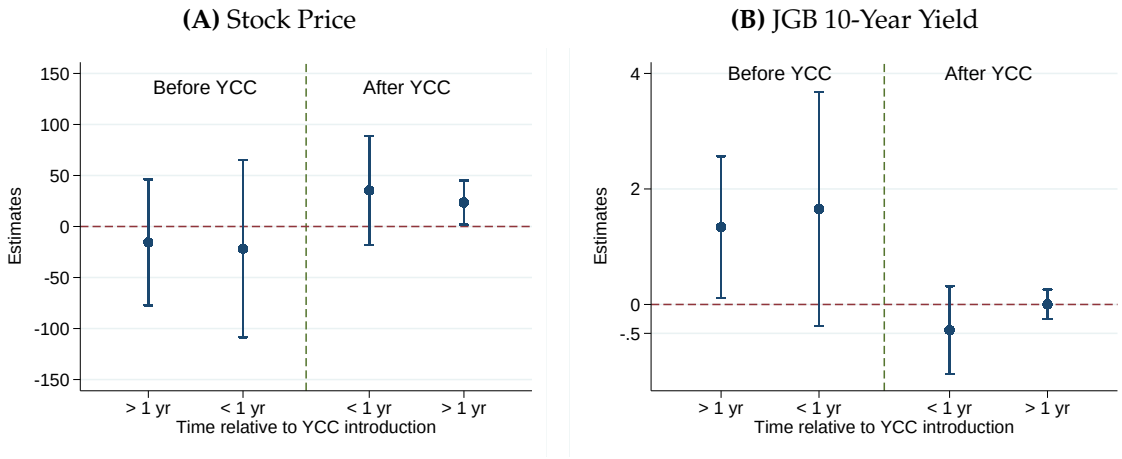


Figure 4: Heterogeneity Around the Introduction of Yield Curve Control

Notes: Panels A and B show, respectively, the responses of stock prices and 10-year JGB yields, divided into two periods relative to the introduction of the YCC on September 21, 2016: (i) more than one year before or after YCC (labeled "> 1yr"), and (ii) within one year of the YCC's introduction (labeled "< 1yr"). The estimates reflect the response from 11 AM on the intervention day to 9 AM the following day. The 90% confidence intervals account for heteroskedasticity and autocorrelation.

long-term rates stopped responding and the effect is precisely estimated as zero. These results imply that yield curve control was successful in stabilizing the long-term interest rate.

One potential concern here is that the observed heterogeneity may not reflect the causal impact of yield curve control. For example, it is possible that the effects of the BoJ's policy intervention gradually increased or decreased over time, irrespective of the implementation of yield curve control. To address this concern, we estimate the effects of the BoJ's stock purchases by focusing on the one-year periods immediately before and after the introduction of yield curve control.

Figures 4A and 4B plot the estimated next-day responses of the stock price and long-term interest rates, dividing the periods into four subsample periods: more than one year before, less than one year before, less than one year after, and more than one year after the introduction of yield curve control. We continue to find the stark heterogeneity even if we focus on the periods shortly before and after the implementation of yield curve control. While standard errors are large, these results support the notion that the heterogeneity reflects the causal effect of yield curve control.[7]

[7]Appendix Figure B.4 further disaggregates the sample into seven finer calendar periods. While standard errors are large, we find that the structural break appears at the time of yield curve control.

4.3 Robustness

Table 1 reports a battery of robustness checks. Rows 1-2 vary the bandwidth, with Appendix Figure B.5 exploring bandwidth choices more systematically. Row 3 uses a quadratic local polynomial, allowing for a more flexible relationship between the running variable and the outcomes. Row 4 controls for BoJ purchases over the previous two days, addressing the concern that our estimates may partly reflect earlier interventions. The estimates are virtually unchanged, consistent with Appendix A.4, which shows that falling below the cutoff today does not predict purchases on other days. Rows 5-6 control for stock returns and long-rate changes over the previous two days, addressing the concern that recent stock- or bond-market conditions may predict subsequent asset-price movements. Finally, row 7 drops observations within one week of cutoff changes, addressing the concern that these changes may be endogenous to economic conditions or signal future BoJ policy. In all cases, the estimates are similar to the baseline estimates.

4.4 Discussion

We close our empirical analysis by discussing several potential concerns. First, Appendix C reports placebo tests at cutoffs that should not trigger BoJ purchases. The placebo estimates are generally indistinguishable from zero and, when statistically significant, have the opposite sign from the baseline estimates. This supports the interpretation that our results are driven by the BoJ policy rule rather than by other factors that may cause discontinuities at arbitrary cutoffs.

Second, if the BoJ is buying or selling long-term government bonds at the same time as the stock purchases, then it is not surprising that the long-term interest rate systematically reacts in response to the stock purchases. On the sales side, during our sample period the BoJ sold government bonds only on March 2017, and March 2020, both after yield curve control. Therefore, BoJ bond sales cannot explain the rise in the long-term interest rate before YCC. On the purchase side, while the BoJ regularly purchased JGBs during our sample period, Appendix D shows no statistically significant discontinuity in JGB purchases, either before or after YCC.

Third, while our results so far concern nominal stock prices and interest rates, one may be concerned that our results are driven entirely by inflation expectations. We show in Appendix Figure B.7 that the response of the inflation swap rate is indistinguishable from zero.

Fourth, the interest rate response is not specific to the 10-year JGB yield but is widespread across the yield curve. In Appendix E, we show that before yield curve control, yields rose

Panel A. Stock Price Response

	All		Before YCC		After YCC	
	Same Day	Next Day	Same Day	Next Day	Same Day	Next Day
0. Baseline	34.90	8.65	36.31	-16.95	37.99	25.01
	(5.89)	(10.18)	(17.50)	(26.27)	(5.13)	(11.62)
1. Narrower Bandwidth	41.65	16.22	30.14	-21.25	49.71	34.67
	(7.16)	(16.09)	(22.18)	(39.57)	(7.65)	(16.90)
2. Wider Bandwidth	29.49	6.07	30.63	-2.44	31.86	18.25
	(4.84)	(9.36)	(13.46)	(20.74)	(4.85)	(9.35)
3. Polynomial Order 2	43.40	16.98	41.13	-22.70	49.37	32.87
	(7.47)	(15.64)	(21.64)	(42.42)	(7.36)	(14.86)
4. Control Past Interventions	35.68	9.54	37.44	-16.80	38.95	26.20
	(5.94)	(10.21)	(17.75)	(26.38)	(5.19)	(11.68)
5. Control Past Stock Returns	34.91	8.62	36.15	-17.33	38.02	23.46
	(5.89)	(10.20)	(17.70)	(26.21)	(5.21)	(12.68)
6. Control Past 10-Year Yield	35.96	6.63	40.91	-23.41	37.72	24.56
	(5.84)	(9.94)	(18.33)	(25.42)	(5.11)	(11.63)
7. Drop Around the Cutoff Changes	34.11	8.11	28.18	-23.75	36.98	29.35
	(6.15)	(11.27)	(16.21)	(28.83)	(5.18)	(12.91)

Panel B. JGB 10-Year Yield Response

	All		Before YCC		After YCC	
	Same Day	Next Day	Same Day	Next Day	Same Day	Next Day
0. Baseline	0.47	0.37	1.52	1.41	0.07	-0.00
	(0.18)	(0.23)	(0.59)	(0.68)	(0.08)	(0.13)
1. Narrower Bandwidth	0.54	0.43	1.96	1.96	0.02	-0.13
	(0.22)	(0.30)	(0.84)	(0.96)	(0.09)	(0.20)
2. Wider Bandwidth	0.40	0.35	1.28	1.05	0.04	-0.03
	(0.16)	(0.19)	(0.47)	(0.50)	(0.07)	(0.10)
3. Polynomial Order 2	0.52	0.43	1.81	1.75	0.10	0.01
	(0.22)	(0.27)	(0.77)	(0.91)	(0.10)	(0.18)
4. Control Past Interventions	0.47	0.37	1.57	1.45	0.07	-0.00
	(0.17)	(0.23)	(0.59)	(0.68)	(0.08)	(0.14)
5. Control Past Stock Returns	0.47	0.37	1.53	1.43	0.06	-0.03
	(0.18)	(0.23)	(0.59)	(0.68)	(0.09)	(0.15)
6. Control Past 10-Year Yield	0.46	0.37	1.51	1.42	0.07	-0.01
	(0.17)	(0.23)	(0.60)	(0.69)	(0.08)	(0.14)
7. Drop Around the Cutoff Changes	0.37	0.28	1.12	1.00	0.05	-0.04
	(0.17)	(0.22)	(0.50)	(0.54)	(0.08)	(0.14)

Table 1: Robustness

Notes: Table 1 shows robustness checks against various modifications of our benchmark specifications. Panels A and B show the responses of stock prices and the JGB 10-year yield, respectively. In each panel, row 0 reports the baseline estimates. Row 1 considers a bandwidth that is 50% less than the original one. Row 2 considers a bandwidth that is 50% larger than the original one. Row 3 considers a local polynomial regression of order 2 instead of 1. Row 4 controls for the BoJ's stock purchases over the past two days. Row 5 controls for stock market returns over the past two days. Row 6 controls for changes in the 10-year yield over the past two days. Row 7 drops observations within one week before and after the cutoff changes. The same-day response indicates the changes in the outcome variable from 11AM to 3PM on the day of the intervention. The next-day response indicates the changes in the outcome variable from 11AM on the day of the intervention to 9AM on the next day. Standard errors, which account for heteroskedasticity and autocorrelation, are reported in parentheses.

at all maturities with larger effects at longer maturities, while after yield curve control all maturities stopped responding.

The final issue concerns the interpretation of our results. One may be concerned that our results are driven by the signaling channel, whereby the BoJ's stock purchases send signals about the BoJ's future policy stance. However, the BoJ announces the target amount of stock purchases each year in advance. Therefore, whether or not the BoJ purchases stocks today should not reveal the BoJ's future policy stance.

5 Interpretations

We now turn to the interpretation of our findings. We argue that our results are consistent with models of inelastic asset demand in which the scope for interest-rate adjustment determines whether stock purchases raise stock prices or interest rates.

Our results are inconsistent with the "inelastic market hypothesis" proposed by Gabaix and Koijen (2021) in its original form. They argue that the demand for stocks is inelastic due to portfolio adjustment frictions so that an inflow into the stock market has a large impact on the stock price. However, since their model features an exogenous interest rate, it predicts that the interest rate is unchanged in response to the central bank's stock purchases, contrary to what we see in the data before YCC.

Instead, our empirical results are consistent with recent work that extends Gabaix and Koijen (2021) by endogenizing the interest rate (Duarte et al., 2025; Caballero et al., 2024). Duarte et al. (2025) show that, even when asset demand is arbitrarily inelastic with respect to return differences, a flow into the stock market has no impact on the stock price but raises the interest rate, exactly as we find in the data before YCC. This is also the prediction from the flexible-price equilibrium version of Caballero et al. (2024).

In these models, an inflow into the stock market puts upward pressure on the stock price. If stock prices were to rise, this would induce a stock market wealth effect on consumption.[8] However, the rise in consumption is not compatible with goods market clearing when output is exogenous. In fact, any movement in the stock price would violate goods market clearing, and this is why the stock price remains unchanged.

If the stock price does not respond, then what equilibrates the excess demand for stocks induced by the central bank's stock purchases? The answer is that the interest rate needs to rise to clear the stock market. As emphasized by Duarte et al. (2025), the argument here does not depend on the degree of portfolio adjustment frictions. Even

[8] Chodorow-Reich et al. (2021) and Di Maggio et al. (2020) provide evidence of the stock market wealth effect on consumption.

when substantial portfolio adjustment frictions make the demand for stocks arbitrarily inelastic, the stock market appears to be perfectly elastic in general equilibrium. The extent of portfolio adjustment frictions only matters for how much the interest rate needs to rise.

The same models also show that the mechanism reverses when the interest rate is fixed by the central bank: the stock price, rather than the interest rate, adjusts to clear the stock market. This is, again, exactly what we found empirically after YCC. In Caballero et al. (2024), the fixed interest rate is compatible with goods market clearing because output is demand-determined, as in the New Keynesian tradition. There, an inflow into the stock market raises the stock price and consumption. The goods market clears because output booms, and the stock market clears because of the higher stock price. The extent of portfolio adjustment frictions determines how much the stock price needs to rise.

Quantitatively, our empirical results suggest a substantial degree of portfolio adjustment frictions. Under YCC, when the interest rate is kept fixed, an inflow of $1 into the stock market raises aggregate stock market value by around $20–$30 (column 6 of Table 1 Panel A). This is substantially larger than the corresponding number of $5 from Gabaix and Koijen (2021). Likewise, in the absence of YCC, an inflow of 1% of stock market capitalization raises the interest rate by around 1–1.5 percentage points. While we are not aware of comparable estimates in the literature, this is a sizable effect. These two results demonstrate that asset demands are substantially inelastic, as the inelastic market hypothesis suggests, but where these inelasticities show up in the financial market critically depends on the (in)ability of the interest rate to adjust.

Taken together, our findings suggest that a small stock-price response to a stock market inflow need not imply elastic asset demand. Inelastic asset demand can instead manifest itself in a large interest-rate response when interest rates can adjust. When interest rates do not adjust, the stock price has to absorb the inflow into the stock market. In Appendix G, we formalize these arguments using a simple theoretical framework that builds on Duarte et al. (2025) and Caballero et al. (2024).

6 Conclusion

How does a flow into the stock market impact financial markets? To answer this question, we exploit discontinuities in the Bank of Japan's policy rule. We empirically show that central bank stock purchases have a far-reaching impact on financial markets. Although these results reject the inelastic market hypothesis in its original form, they give strong support to its general equilibrium extension with endogenous interest rate adjustment.

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